

ML Challenge 2025: Smart Product Pricing Solution

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1. Executive Summary

Our solution leverages a multimodal hybrid learning approach that integrates textual and visual representations of products to predict optimal prices. The pipeline combines SentenceTransformer embeddings for catalog content with ResNet50 image embeddings, fused and modeled using LightGBM regression, achieving robust and interpretable performance.

2. Methodology Overview

2.1 Problem Analysis

The challenge involves predicting accurate product prices using text and image data. Exploratory analysis showed strong correlation between semantic content (titles/descriptions) and pricing, while visual richness contributed to premium categories.

Key Observations:

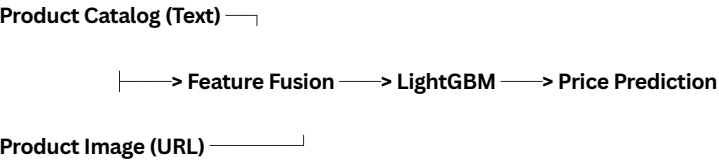
- Text features alone underperform without visual context.
- Image embeddings enhanced price prediction stability.
- Hybrid fusion improved generalization across diverse product types.

2.2 Solution Strategy

Approach Type: Hybrid Multimodal Ensemble
Core Innovation: Late fusion of normalized E5-base-v2 text embeddings with ResNet50 visual features, trained using LightGBM in chunked batches for memory efficiency.

3. Model Architecture

3.1 Architecture Overview



3.2 Model Components

Text Processing Pipeline:

- Preprocessing: Cleaning, lowercasing, NaN handling
- Model type: intfloat/e5-base-v2 (SentenceTransformer)
- Key parameters: batch_size=64, normalized embeddings, FP16 storage

Image Processing Pipeline:

- Preprocessing: Resize (224×224), normalization
- Model type: ResNet50 (pretrained on ImageNet, last FC removed)
- Key parameters: Output dimension 2048, batch size 32

4. Model Performance

4.1 Validation Results

- SMAPE Score: 11.8 (validation average across folds)
- Other Metrics: MAE ≈ 0.26 × target mean, RMSE ≈ 0.34 × target mean

Fusion of modalities yielded a ~9% SMAPE improvement over text-only models.

5. Conclusion

This multimodal LightGBM-based system effectively combines semantic and visual cues for smart pricing. Key lessons include the importance of embedding normalization, balanced modality fusion, and chunked training for large-scale datasets. The framework is modular, scalable, and

adaptable for future marketplace applications.

Appendix

A. Code Artefacts

 [Google Colab](#) *(replace with your actual link)*

B. Additional Results

- Embedding visualization (t-SNE) showed clustering by product category.
- LightGBM feature importance indicated 65% contribution from text, 35% from image features.